Minimization of Objective function to obtain BEST model parameters

$$J(\underline{W}) = \tfrac{1}{2}(Y-\hat{Y})^T(Y-\hat{Y}) + \tfrac{1}{2}\lambda \underline{W}^T\underline{W} = \tfrac{1}{2}(Y-X\underline{W})^T(Y-X\underline{W}) + \tfrac{1}{2}\lambda \underline{W}^T\underline{W}$$

$J(\underline{W})$ $+\tfrac{1}{2}\lambda \underline{W}^T\underline{W}$ $+\tfrac{1}{2}\lambda \underline{W}^T\underline{W}$

$$J(\underline{W}) = \tfrac{1}{2}[Y^TY + (X\underline{W})^T(X\underline{W}) - Y^T(X\underline{W}) - (X\underline{W})^TY] + \tfrac{1}{2}\lambda \underline{W}^T\underline{W}$$

$J(\underline{W})$ $+\tfrac{1}{2}\lambda \underline{W}^T\underline{W}$

$$J(\underline{W}) = \tfrac{1}{2}[Y^TY + (X\underline{W})^T(X\underline{W}) - 2(X\underline{W})^TY] + \tfrac{1}{2}\lambda \underline{W}^T\underline{W}$$

$J(\underline{W})$ $+\tfrac{1}{2}\lambda \underline{W}^T\underline{W}$

$\frac{\partial J(\underline{W})}{\partial \underline{W}} = 0$ Get Best W that minimizes the Objective Function J(W)

Remember

$\frac{\partial \underline{W}^T\underline{W}}{\partial \underline{W}} = 2\underline{W}$

$$\frac{\partial J(\underline{W})}{\partial \underline{W}} = 0 = \tfrac{1}{2}[(X\underline{W})^T(X) + (X)^T(X\underline{W}) - 2(X)^TY] + \lambda \underline{W}$$

$\frac{\partial J(\underline{W})}{\partial \underline{W}}$ $J(\underline{W})$ $+\lambda \underline{W}$

$$\frac{\partial J(\underline{W})}{\partial \underline{W}} = 0 = \tfrac{1}{2}[2X^TX\underline{W} - 2X^TY] + \lambda I\underline{W}$$

$\frac{\partial J(\underline{W})}{\partial \underline{W}}$ $J(\underline{W})$ $+\lambda I\underline{W}$

$$(X^TX + \lambda I)\underline{W} = X^TY$$

λI

$$\underline{W} = (X^TX + \lambda I)^{-1}X^TY$$

λI